NAME: ______________________ **PERIOD:** ________ **DATE:** ____________

When a Normal Curve Distorts the Tail

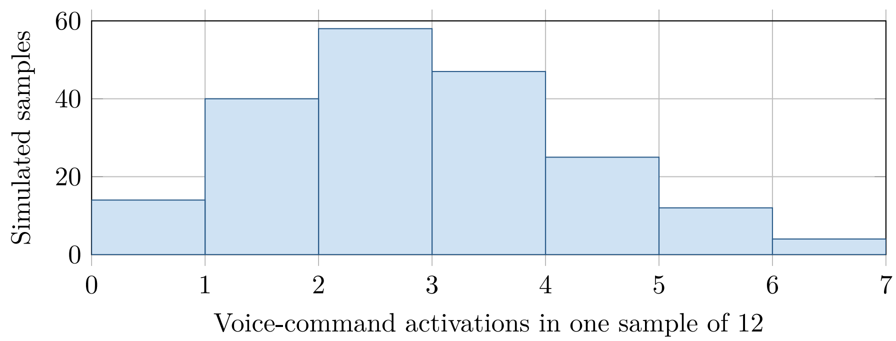
AP Statistics · Topic 6.1 (CED 3.3) · B: 2026-11-24 · E: 2026-12-23

[STAR] TODAY'S PROBLEM

Suppose an app developer's model says that $p_0 = 0.20$ of all new users activate voice commands during their first day. A simple random sample of 12 new users includes 6 who activate the feature, so $\hat{p} = 0.50$. Under the developer's model, a class simulates 200 samples of 12 independent users; the histogram shows the number of activations in each sample.

Mara: "A normal model for $\hat{p}$ gives $P(\hat{p} \geq 0.50) \approx 0.005$, so that is the chance probability I would report."

Jules: "The simulated distribution is not very normal. I would estimate the chance probability directly from its tail."



FIRST TAKE — before the video (5 min)

Which chance probability would you report, Mara's or Jules's? Cite one detail from the histogram or sample size.

[BOOK] CHECK THE SHAPE BEFORE USING NORMAL (keep on the page)

Under a proposed proportion p_0, the sampling distribution of $\hat{p}$ has center p_0 and standard deviation $\sqrt{p_0(1-p_0)/n}$. A normal approximation needs both expected counts $np_0 \geq 10$ and $n(1-p_0) \geq 10$. A matched simulation does not need a normal shape: estimate a one-sided chance probability by counting simulated results at least as large as the observed result. A small chance probability is evidence against the proposed model, not proof of a particular cause.

Finish after the video:

DOK 1 **(a)** State the most common number of activations in the simulated samples and describe the distribution as roughly symmetric, skewed left, or skewed right.

DOK 2 **(b)** Use the histogram to estimate $P(\hat{p} \geq 0.50)$ under $p_0 = 0.20$. Then calculate np_0 and $n(1-p_0)$ and assess the expected-count condition for a normal approximation.

DOK 3 **(c)** Choose the better chance-probability method and use a 5% criterion to judge evidence against $p_0 = 0.20$. Cite the simulated tail and a shape or expected-count feature, compare 0.020 with 0.005, and explain the reader consequence of Mara's report.

Frame: I would use ____________ because the expected counts are _______________ and the simulated shape is ______________.

Frame: The tail estimate ________ means _______________________; reporting 0.005 instead would make a reader think ______________________.

EXIT — turn this in

One thing the video changed about my first take: __________________________________